

Structured Illumination Microscopy (continued)

Structured illumination microscopy (SIM) is the first of three highly utilized super-resolution techniques we'll cover in class

Microscopists generally prefer uniform Kohler illumination, but there are exceptions.

SIM beats the diffraction limit by a factor of two using structured, as opposed to uniform, illumination

SIM's mathematical foundations are complicated, but the key points can be explored via two analogies

(1) Analogy w/ moiré patterns

Big idea: Microscopists want to know the distribution of cellular constituents

Assume that distribution of fluorophores mirrors the distribution of cellular constituents

The camera (or other detector) measures fluorescence F , not the underlying distribution, ρ , of fluorophores.

The quantities F & ρ are related as follows:

$$\boxed{F \propto \rho \times I} \quad \left\{ \begin{array}{l} F = \text{measured fluorescence} \\ \rho = \text{distribution or density of fluorophores} \\ I = \text{intensity of excitation light} \end{array} \right.$$

$$\left. \begin{array}{l} \text{Measure } F \\ \text{Know } I \end{array} \right\} \Rightarrow \text{Calculate } \rho$$

This leads to a factor of two improvement in resolution

Why does this work?

Understand conceptually via moiré patterns

Moiré concept $\left\{ \begin{array}{l} \text{Superimpose two closely spaced patterns} \\ \text{Get a more widely spaced pattern} \end{array} \right.$

- (1) Assume a cellular spacing that's too close to be resolved ($< 200 \text{ nm}$)
- (2) Superimpose illumination pattern at resolution limit

(3) This will create a moiré pattern that can be resolved because it's $\geq 200 \text{ nm}$

Reconstruct ρ from measured F to better than the diffraction limit in all three dimensions

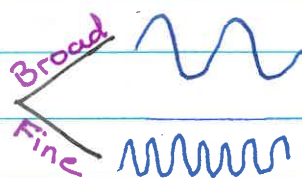
Beats

(2) Analogy w/ Beats

Started w/ look at MRI because it is a frequency-based technique

An image is a sum of information from many spatial frequencies

Roles of
different
frequencies



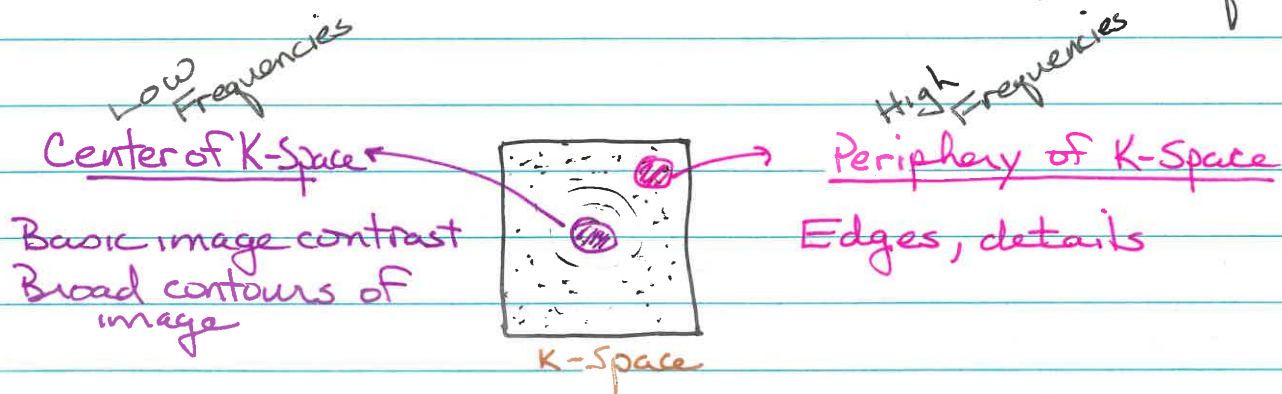
Low frequencies encode big global features in image

High frequencies encode fine details (e.g., edges) in image

ANIMATION

Questions & Answers
in MRI

Location of
Spatial Frequencies



The quest for higher resolution in imaging is a quest for collecting higher spatial-frequency information

The goal is to enhance the range of collected frequencies w/o losing the low frequencies.

Collect fewer frequencies $\Rightarrow$ resolution will be poorer

Beats concept { Adding together two sounds (waves) of slightly different frequencies yields a lower-frequency beating sound

The beat frequency is the difference in the frequencies of the two contributing waves

$$\boxed{\nu_{\text{beat}} = \nu_2 - \nu_1} \quad \left\{ \begin{array}{l} \nu_{\text{beat}} = \text{beat frequency} \\ \nu_2 = \text{higher frequency} \\ \nu_1 = \text{lower frequency} \end{array} \right.$$

The beat frequency, ν_{beat} , is lower than the two contributing frequencies, ν_2 & ν_1 , because it's a difference between the two

Example: $\nu_2 = 110 \text{ Hz}$
 $\nu_1 = 100 \text{ Hz}$
 $\Rightarrow \nu_{\text{beat}} = 10 \text{ Hz}$

Connection w/ structured illumination:

$$\nu_2 = \nu_{\text{object}}$$

Frequency we want to resolve in image

$$\nu_1 = \nu_{\text{illumination}}$$

Diffraction-limited frequency of excitation light

$$\nu_{\text{beat}} = \nu_{\text{fluorescence}}$$

Diffraction-limited frequency of measured fluorescence (max resolvable ν in image)

Rewrite definition of beats using these terms
 & solve for ν_{obj}

$$\nu_{Fluorescence} = \nu_{Object} - \nu_{Illumination}$$

$$\Rightarrow \nu_{Object} = \nu_{Fluorescence} + \nu_{Illumination}$$

$$= \nu_{Diffraction\ Limit} + \nu_{Diffraction\ Limit}$$

$$\boxed{\nu_{Object} = 2 \cdot \nu_{Diffraction\ Limit}}$$

Thus, in this analogy, by using diffraction-limited excitation light, & by measuring fluorescence in the image at the diffraction limit, you can derive information in the object distribution at twice the diffraction limit (where twice the frequency in frequency space corresponds to resolving features at one-half the spacing in object space)

Implementation

- Use grating to generate structured illumination
- Get "striped" images
- Use multiple striped images at different angles to reconstruct image w/ 2x resolution using computer

SIM strengths:

- (1) Uses standard fluorophores
- (2) Fast
- (3) Straightforward multi-color imaging

SIM weaknesses:

- (1) Modest (2-fold) resolution enhancement
- (2) Requires post-acquisition processing
(artifacts, calibration issues)